Quiz-1 prep (Lec 2, 3, 4, 5) + Lec 1

Set-A

1.	How many different colors you can have in a 3 bit/pixel image? How many bytes are required to store an image of 20x20 having 64 different intensity levels?	1

→ Here, given $K = 3$ bit/pixel

∴ we can have no of different colors $= \boxed{L = 2^k} = 2^3 = 8$ (Ans)

Again,

For 20X20 image, $M = 20$, $N = 20$, $L = 64$ (diff intensity levels)

we know that,

$L = 2^k$

$\Rightarrow 64 = 2^k$

$\Rightarrow K = \frac{\log 64}{\log 2} = 6$

∴ The no of bit size, $\boxed{b = M * N * K}$

$= (20 * 20 * 6)$ bits $= 2400$ bits

∴ no of bytes to store 20X20 img $= \frac{2400}{8}$ bytes ; [∵ $\boxed{1 \text{ byte} = 8 \text{ bits}}$]

$= 300$ bytes (Ans)

2.	Consider the 8-bit image of fig.3(a) and apply the transformation function of fig.3(b). on it. Show the matrix of new image and its histogram.	2

4	4	8	8
2	4	4	2
2	4	4	2
10	15	15	10

Fig.3(a)

$S = C X r^2$

Assume c= 1

Fig.3(b)

→ After applying, $S = cr^2$; where $c = 1$,
The output matrix becomes like this. This is the new img matrix after transformation:

16	16	64	64
4	16	16	4
4	16	16	4
100	225	225	100

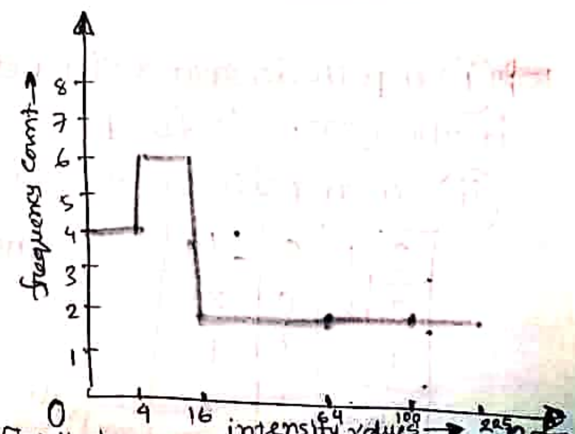


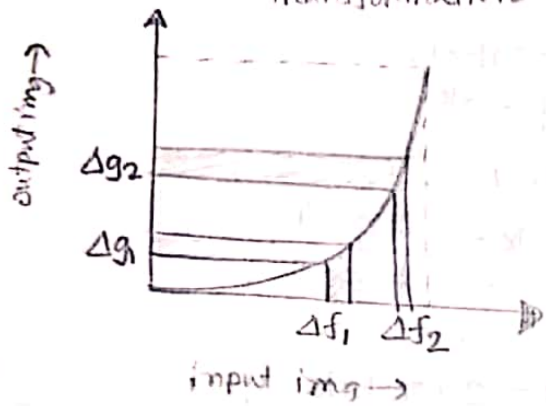
Fig: Histogram

3

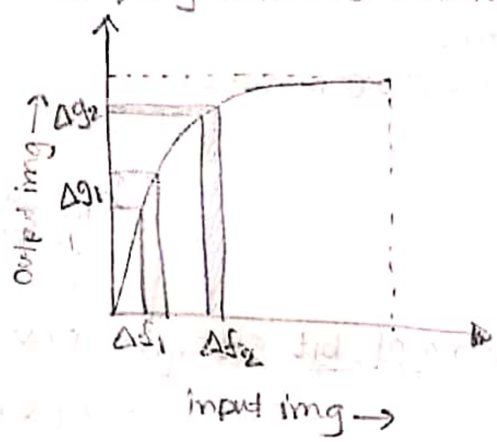
Original Image | Transformation A | Transformation B

Show the possible 'general' shape of the transformation functions A and B

A ⇒ inverse logarithmic Transformation



B ⇒ Logarithmic Transformation



4 marks Spring 18

4

Consider the following image segment:

3	1	2	1(a)
2	2	0	2
1	2	1	1
1(b)	0	1	2

Calculate D_4, D_8, D_e = ?

V = {0,1}

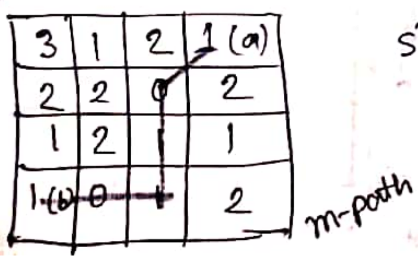
i) Does a 4 – path exists between a and b.

ii) Does a m – path exists between a and b.

(If your answer is yes then draw the shortest path and calculate the length of the path otherwise, explain why a particular path does not exist)

(i) 4-path doesnot exist betn a and b. Because, there is no $N_4(a)$ in the given V = {0, 1}

(ii) For m-path, we need to know the m-adjacency of neighbors,

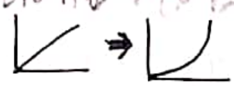


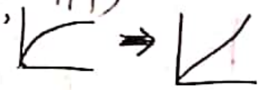
shortest m-path length = 5

Gamma Correction: A variety of devices (img capture, printing, display) respond according to a power law. By convention, the exponent in the power law eqn, $s = cr^{\gamma}$: is referred to as Gamma(γ). The process used to correct the power law response phenomena is called ....

It is important if displaying an image accurately on a computer screen is of concern. Images that are not corrected properly can either look bleached out or too dark. Trying to reproduce colors accurately also requires some knowledge of γ correction, because varying the γ value changes not only on the basis of several experiments.

(γ correction মানে হচ্ছে যে device এ print করছি, তাদের একটি unique power law আছে। যদি তারা যে law follow করে তার ওইটার inverse করে দেয়া যায় সেই অনুযায়ী, then স্ক্রীনে আউটপুট ঠিক দেখাবে, কোনো prob হবে না।)

Before γ correction, 

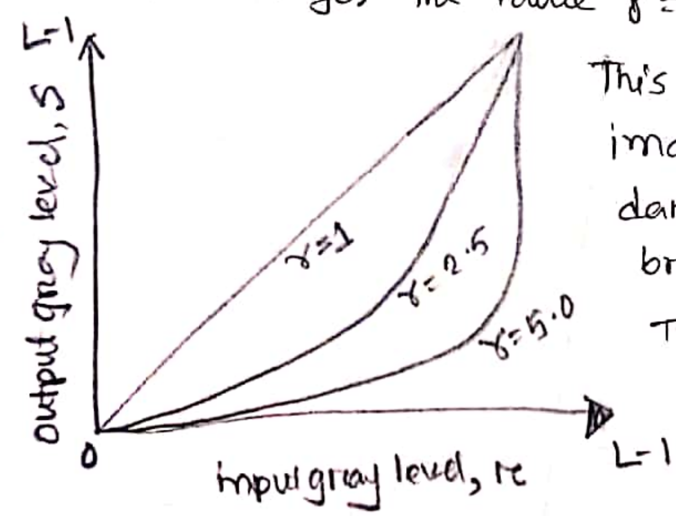
After γ correction, 

By default, the software I made it uses $\gamma = 2.5$

We know the power law transformation formula is:

$$s = cr^{\gamma} = T(r)$$

If the user changes the value $\gamma = 5.0$, the value of s will increase.



This means the curve for the output image will ~~increase~~ compress dark pixel values ~~more~~ and expand brighter pixels more than default. The output image gets high contrast, some regions become too dark.

Set-B

1 Suppose "lena.tif" is a square image with 8bit gray values whose pixels size is 256. Calculate gray level resolution and spatial resolution of "lena.tif". Also calculate the bit depth and bitsize of lena.tif.

⇒ Here, $k = 8$ → bit depth

Gray level Resolution of lena.tif, $L = 2^k$
$= 2^8 = 256$ levels

Spatial Resolution = ?

Here, bitsize of the image $= M^2 * K$; [∵ pixelsize $= M \times N = M^2 = 256$]
$= (256 * 8)$
$= 2048$ bits

2 Consider the 8-bit image of fig 3(a) and apply the transformation func fig 3(b) on it. Show the matrix of new img and its histogram.

16	16	64	64
4	16	16	4
4	16	16	4
100	225	225	100

fig 3(a)

$S = C \times r^{\frac{1}{2}}$

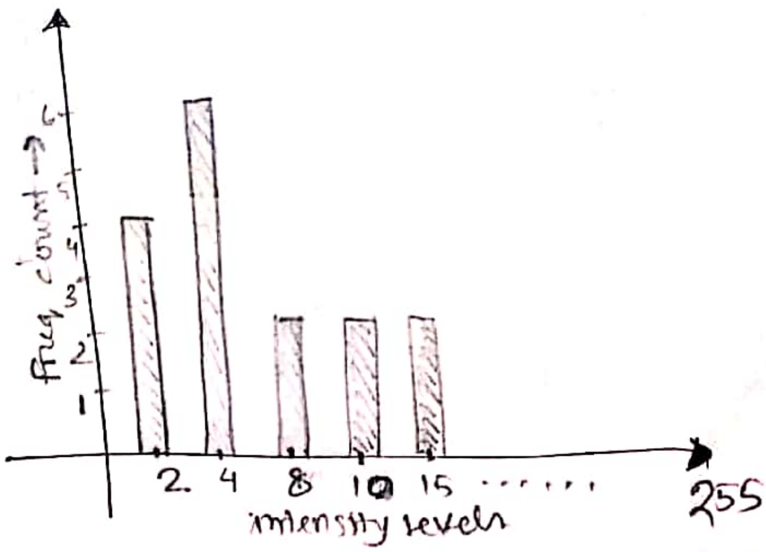
Assume, $c = 1$

fig: 3(b)

⇒

4	4	8	8
2	4	4	2
2	4	4	2
10	15	15	10

New img matrix after transformation



Spring 18 · 2 marks

3. One of the simplest piecewise linear functions is a contrast-stretching transformation, where the locations of points $(r1, s1)$ and $(r2, s2)$ control the shape of transformation func. Now illustrate the relation betn $(r1, s1)$ and $(r2, s2)$ for : (i) Thresholding
(ii) Contrast stretching

→

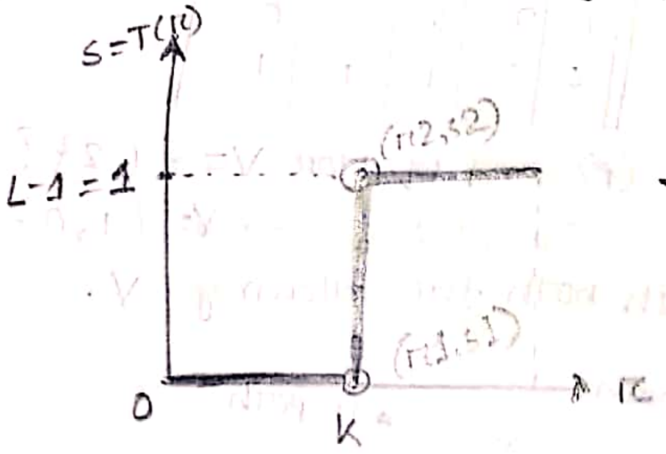
(i) Thresholding : In this linear, gray level transformation point processing technique, we have to select a threshold point k.

$$S = T(r) = \begin{cases} 1 & ; r > \text{threshold} \longrightarrow \text{white} \\ 0 & ; r \leq \text{threshold} \longrightarrow \text{black} \end{cases}$$

Firstly, we will select a threshold point k from the background of an image.

- replacing the values below k, to 0 (black)
- " " above k to 1 (white).

As a result, we will get a binary img.

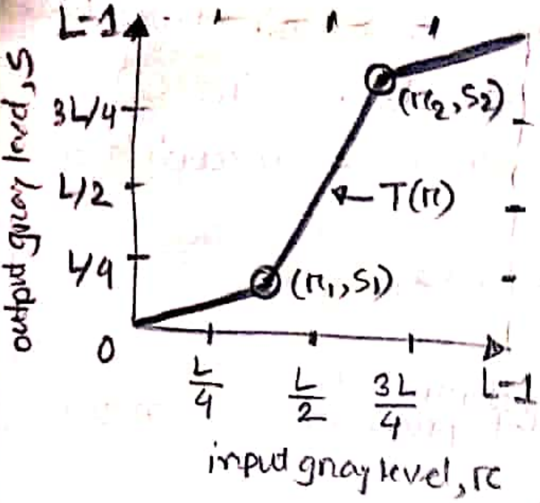


if ~~(r1 = r2) then s1 = 0 and s2 = L-1~~

- If $r_1 = s_1$ and $r_2 = s_2$; there is no change in gray levels
- If $r_1 = r_2$, $s_1 = 0$ & $s_2 = L-1$; then it is a threshold function. The resulting img is binary.

(ii) Contrast Stretching / Normalization : It is a process that expands the range of intensity levels in an img so that it spans the intensity range of the recording medium or display device.

(input img এর highest ratio number দিয়ে multiply করে difference দিয়ে divide করা)



This figure shows a typical transformation used for contrast stretching. The location of points (r_1, s_1) and (r_2, s_2) control the shape of the transformation func. Here,

$r_1 = r_{min}$ & $s_1 = 0$

$r_2 = r_{max}$ & $s_2 = L-1$

- min and max intensity range বেশি → contrast বেশি
- " " " " " কম → " কম

4 Consider the two img subsets S1 and S2:

S1					S2				
0	0	0	3	1	2	1(q)	1	1	0
1	0	0	2	2	0	2	0	0	1
1	0	0	1	2	1	1	0	0	0
0	0	1	(p) 1	0	1	2	0	0	0
0	0	1	1	1	0	0	1	1	1

(i) Does 4-path exists betn (p) and (q) for $V=\{1,2\}$?
(ii) Do m-path " - - - - - - - - - - $V=\{1,0\}$?

exists 4-path & m-path both for following V.

4path

m-path

S_1					S_2				
0	0	0	3	1	2	1(q)	1	1	0
1	0	0	2	2	0	2	0	0	1
1	0	0	1	2	1	1	0	0	0
0	0	1	(p) 1	0	1	2	0	0	0
0	0	1	1	1	0	0	1	1	1

(or 2, 3, 4, 5)

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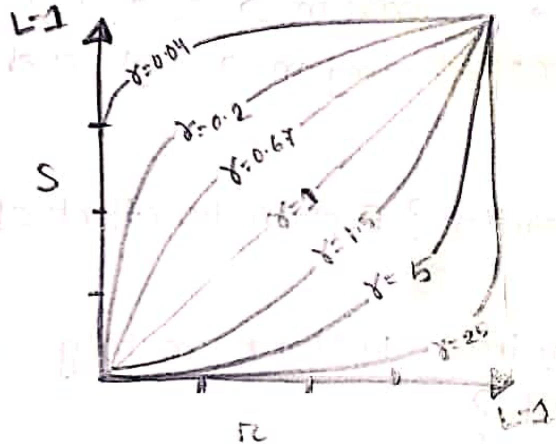
5 For a given image, using power law transformation function, what will be the effect on output img? (i) $\gamma > 1$
(ii) $\gamma < 1$
(iii) $\gamma = 1$

➡ Power law transformation have the basic form:

$$s = cr^{\gamma}$$

Here, c and γ are some (+ve) constants. In the following figure, we notice a family of possible transformation curves obtained by varying γ.



we can see that, the curves generated with values of $\gamma > 1$ have exactly opposite effect as those generated with values of $\gamma < 1$. The eq^n reduces to the identity transformation when $c = \gamma = 1$. So, we can say, using this P.L.T func, the effect on the output img in following conditions are:

(i) $\gamma > 1 \Rightarrow$ ~~Exp~~ compresses values of dark pixels
expands " " brighter "

(ii) $\gamma < 1 \Rightarrow$ opp.

(iii) $\gamma = 1$ and $c = 1 \Rightarrow$ Input and output img will be identical, there is no change.
$(s = r)$

Spring 18

2 marks

$\boxed{3}$ (a) What is digital image processing? DIP key stages?

➡ DIP : it focuses on 2 major tasks :
(1) improvement of pictorial info for human interpretation
(2) processing of img data for storage, transmission and representation for autonomous machine perception.

- img processing : img in → img out
- img analysis : img in → measurements out
- img understanding : img in → high level description out

4 marks

$\boxed{3}$ (c) What is spatial resolution? Explain the effect of ~~an~~ spatial resolution of an img? (slide)
How many bytes are required to store an img of 20X20 having 64 different intensity levels?

➡

M=20 N=20 L=64	$L=2^k$ $\Rightarrow 64=2^k$ $\Rightarrow k = \frac{\log 64}{\log 2} = 6$

$\therefore b = M * N * K$
$= (20 * 20 * 6)$
$= 2400$ bits
$= \frac{2400}{8}$ bytes $= 300$ bytes.

3 (a) Describe the RGB color model with schematic of RGB color cube.

→

4 (a) What is Image Enhancement?

→ processing an img so that the result is more suitable than the original img for a specific application. ultimate purposes:

i) highlighting interesting detail in imgs
ii) removing noise from imgs
iii) making imgs more visually appealing

2 methods:

(i) Spatial domain → direct manipulation of pixels in an img

(ii) Freq. " → process the img by modifying the fourier transform of an img

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4 marks

$\boxed{1}$ (a) (i) "A digital img is always only an approximation of real world scene" — why? Mention 2 techniques those made this digital representation of analog world possible. ?

(ii) What is the basic differences betn RGB img and indexed color img? which representation is preferred in case of storage, transmission and rendering a large size img?

⇒

4 m

(i) Digitization implies that a digital img is an approximation of a real world scene.

(ii)

RGB Image	Indexed Image
(1) Each pixel contains a vector representing Red, Green and Blue components.	(1) Each pixel contains index number pointing to a color in a color table.
(2) In modern systems, RGB color has become the norm.	(2) Advantage: lower memory requirements (img size ↓), easier to supported in Hardware Disadv: limited color available on displays

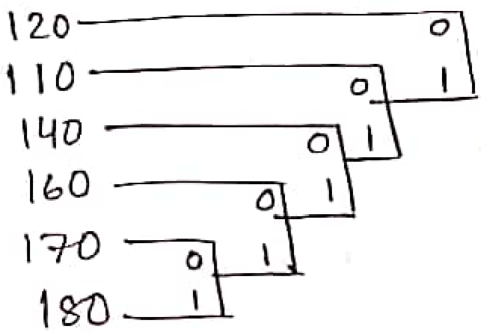
$$\therefore \tilde{H} = -\{0.04(\log_2 0.04) + 0.16(\log_2 0.16) + 0.2(\log_2 0.2) + 0.28(\log_2 0.28) + 0.2(\log_2 0.2) + 0.12(\log_2 0.12)\}$$

$$= -\{-0.185 - 0.423 - 0.464 - 0.51 - 0.464 - \cancel{3}\,0.367\}$$

$$= 2.413 \text{ bits/pixel} \quad \text{(Ans)}$$

(ii) Sorting the table :

120 → 7
110 → 5
140 → 5
160 → 4
170 → 3
180 → 1



codeword
0
10
110
1110
11110
11111

(iii)

← no of bits to represent r_k using Huffman

r_k	$l(r_k)$	$P(r_k)$
120	1	0.28
110	2	0.2
140	3	0.2
160	4	0.16
170	5	0.12
180	5	0.04

$$L_{avg} = (1 * 0.28) + (2 * 0.2) + (3 * 0.2) + (4 \times 0.16) + (5 * 0.12) + (5 \times 0.04)$$

$$= \cancel{2.43} \; 2.72$$

$$C_R = \cancel{2.72} \; \frac{n_1}{n_2} = \frac{8}{2.72} = 2.94$$

$$\therefore R_D = 1 - \frac{1}{C_R} = 1 - \frac{1}{2.94} = 0.659$$

(iv)

The effectiveness of Huffman coding

$$= \boxed{\left(\frac{\tilde{H}}{L_{avg}} * 100\right)}$$

$$= \left(\frac{2.413}{2.72} \times 100\right)\%$$

$$= 88.71\%$$

efficiency

| 1 (0) 0 | 1 | 2 | m-path

Fall 18

2 How many types of Redundancies are there in an image? Give brief description of each.

⟹

- Redundancy
  - Statistical R
    - InterPixel R (corresponding to statistical dependencies among neighboring pixels)
      - Spatial R
      - Temporary R
    - Coding R
      - fixed length coding
      - variable length
        - Huffman
        - Arithmetic
  - Psychovisual R (corresponding to eliminating less relative info in a visual obj)

see slide

Each pixel, 1 byte = 8 bits ⟶ B&W img / Gray Scale
" " 3 bytes = 24 bits ⟶ Color img (one byte each for R, G, B)

$\boxed{1\ GiB = 10^9\ Bytes}$

3 Without using any compression, how many bytes are required to store a 30min Gray Scale video with a frame rate of f and a resolution of m×n pixel?

⟹ ~~For 30min long GrayScale video,~~ For a single second,
The amount of data to be accessed is:

$= f\ \text{frames} * (m * n)\ \frac{\text{pixels}}{\text{frame}} * 1\ \frac{\text{byte}}{\text{pixel}}$ [∵ Grayscale]

$= (f * m * n)\ \frac{\text{bytes}}{\text{sec}}$

∴ For a 30min long Grayscale video, the data is:

$= (f * m * n)\ \frac{\text{bytes}}{\text{sec}} * (30 * 60)\ \text{sec}$

$= 1800\ fmn$ bytes

(Ans)

| 1 | 0 | 1 | 2 | m-path

8130
Spring18

Quiz-4 (Set B)

1. Use ~~tempet~~ (LZW) . . . 4x4, 8bit img 2D array

7	3	23	42
7	3	23	42
7	3	23	42
7	3	23	42

(i) Show your step by step LZW encoding process. To generate your codebook, use 9 bits.

(ii) Any compression achieved by employing LZW in your above encoding process.

(i) ➡

recognized pixel / currently being processed	Pixel being processed	Encoded output	Dictionary location (codeword)	Dictionary entry
–	7	–	–	–
7	3	7	256	7-3
3	23	3	257	3-23
23	42	23	258	23-42
42	7	42	259	42-7
7	3			–
7-3	23	256	260	7-3-23
23	42			–
23-42	7	258	261	23-42-7
7	3			–
7-3	23			–
7-3-23	42	260	262	7-3-23-42
42	7			–
42-7	3	259	263	42-7-3
3	23			–
3-23	42	257	264	3-23-42
42		42		–

(ii) Without using LZW,
we would use 8bit 16pixel value

∴ Original img = (16×8) bits = 128bits = n_1

With LZW, it has became 9bit, 10pixel value

∴ Compressed img = (10×9) bits = 90bits = n_2

$\therefore C_R = \frac{n_1}{n_2} = \frac{128}{90} = 1.42$

∴ 128bits are compressed into ~~9bit~~ 90bits

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3. A 1024×1024 8bit img with 5.3bits/pixel entropy is to be Huffman coding. What is the maximum compression that can be expected?

➡ Here given,
$\tilde{H} = 5.3$ bits/pixel

1	0	0	1	2

m-path

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(5)(b)(ii)

An img of size 10×10 has only 3 colors, 50 pixels in the img are black (0), 30 pixels are white (255) and 20 pixels are gray (120). Using variable length coding, the following codes are assigned to these 3 colors as shown:

color	code
0	1
255	0
120	111

Find the avg no of bits per pixel required to encode this img.

➡ For variable length coding, suppose ~~we can use~~ Huffman coding technique is being used.

Total pixel count = 10×10 = 100

$\therefore P(\text{Black}) = P(0) = \frac{50}{100} = 0.5$

$P(\text{white}) = P(255) = \frac{30}{100} = 0.3$

$P(\text{gray}) = P(120) = \frac{20}{100} = 0.2$

So we can construct the table:

r_K	$l(r_K)$	$P(r_K)$	codeword
0	1	0.5	1
255	1	0.3	0
120	3	0.2	111

$\therefore$ avg no of bits per pixel to encode this img,

$L_{avg} = \sum_{K=0}^{L-1} l(r_K) P(r_K)$

$= (1 * 0.5) + (1 * 0.3) + (3 * 0.2)$

$= 1.4$ bits/pixel

(c) 4×7 pixel img (8bit code for each pixel)

21	21	21	95	16	43	43
21	21	21	95	16	43	43
21	21	21	95	16	43	43
21	21	21	95	16	43	43

(i) Entropy, $\tilde{H} = \sum_{K=0}^{L-1} -P_i(\log_2 P_i)$

$P_{21} = 12/28 = 0.428$

$P_{95} = 4/28 = 0.142$

$P_{16} = 4/28 = 0.142$

$P_{43} = 8/28 = 0.285$

$\therefore \tilde{H} = -\{0.428(\log_2 0.428) + 0.142(\log_2 0.142) + 0.142(\log_2 0.142) + 0.285(\log_2 0.285)\}$

$= -\{-0.524 - 0.39 - 0.39 - 0.516\}$

$= 1.82$ bits/pixel

(v) Huffman coding effectiveness

$= \left(\frac{\tilde{H}}{L_{avg}} * 100\right)\%$

$= \frac{1.82}{1.85} * 100$

$= 98.37\%$ (Ans)

(ii) Sorting the table,

21	12
43	8
95	4
16	4

	codeword
21	0
43	10
95	110
16	111

(iii)

r_K	$l(r_K)$	$P(r_K)$
21	1	0.428
95	3	0.142
16	3	0.142
43	2	0.285

$\therefore L_{avg} = (1 * 0.428) + (3 * 0.142) + (3 * 0.142) + (2 * 0.285)$

$= 1.85$

$\therefore C_R = \frac{8}{1.85} = 4.32$

(iv) $R_D = 1 - \frac{1}{4.32} = 0.768 = 76.85\%$

Scanned by CamScanner

MORPHOLOGY - PART(I)

Maths & Prev Year Ques

Spring-18

1 (a) [2 marks]

What is the difference betn filter mask and structuring Element (SE)? Give Example of box and cross SE.

4 marks

(b) Given the img A in figure (i) and SE B in (ii):

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

A
(i)

1
1
1

B
(ii)

(i) Sketch the result of Erosion of A by B
(ii) - - - - - - - - - Dilation of A by B
(iii) - - - - - - - Closing " A by B
(iv) - - - - - - - Opening " A by B

(i) Erosion

0	0	0	0	0	0
0	0	0	0	0	0
0	0	1	1	0	0
0	0	0	0	0	0
0	0	0	0	0	0

(ii) Dilation

0	0	1	1	0	0
0	1	1	1	1	0
0	1	1	1	1	0
0	1	1	1	1	0
0	0	1	1	0	0

(iii) Closing (Dilation+Erosion)

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

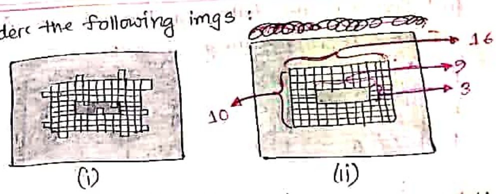
(iv) Opening (Erosion + Dilation)

0	0	0	0	0	0
0	0	1	1	0	0
0	0	1	1	0	0
0	0	1	1	0	0
0	0	0	0	0	0

col = 3+9+4
row = 4+3+3
∴ col = 16
row = 10

4 marks

(c) Consider the following imgs :



(i) (ii)

see ques

Now propose a morphological procedure to clear the edge artifacts of the img given in (i). such that the img in (ii) is obtained. Clearly state the structuring element(s) and no of iterations that you would use in your procedure

The SE will be 3X3 square, which will perform OPENING (erosion ~~after~~ followed by Dilation) on img (i) and thus we will get image (ii). The SE is :

1	1	1
1	1	1
1	1	1

After performing Erosion by SE, the image (i) becomes

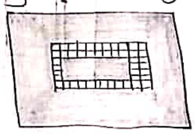


Fig: x

The edge artifacts of the input image is being cleared. But, the foreground has been shrinked a bit.

After performing Dilation by SE on the x image, it becomes exactly same as the img (ii). Here, the foreground is enlarged.

∴ Total iterations $= (16+2) * (10+2) - (3 * 9)$
$= 189$ iterations

1(0)	0	1	2

m-path

4 marks

(d) Consider the following binary img A in figure. Show the result of applying a 3X3 SE.

CUET

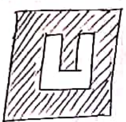
$B = (A \oplus S) \cap A^c$

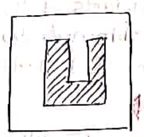
$B = (A \ominus S) \cup A^c$

$B = (A \oplus S) \cup A^c$

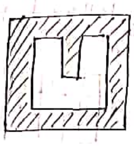
A

Here,

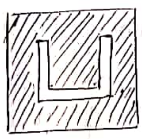
$A =$ 

$\therefore A^c =$ 

$\therefore SE =$ 

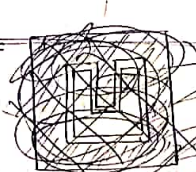
$\therefore A \oplus S =$ Dilation $=$ 

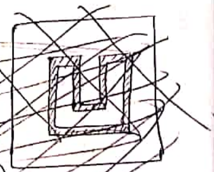
সাদা বাড়ছে

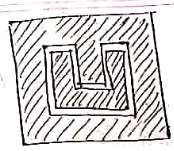
$\therefore A \ominus S =$ Erosion $=$ 

কালো বাড়বে

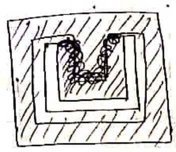
$\therefore$ ~~(i) $B = (A \oplus S) \cap A^c =$~~





(i) $B = (A \oplus S) \cap A^c =$ 

common part নিতে হবে

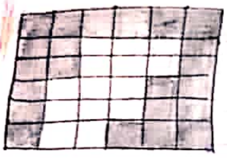
(ii) $B = (A \ominus S) \cup A^c =$ 

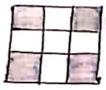
এ ques এ $A \ominus S$?

(iii) $B = (A \oplus S) \cup A^c =$ 

Fall 18

4 (c) Given the img A and SE → S. Considering white pixels are foreground, black are background. Compute the following:





(i) A^c
(ii) (A dilated by S)
(iii) (A eroded by S)
(iv) (A dilated by S) − A
(v) A − (A eroded by S)

Here,

Here, Black = 0
White = 1

0	0	0	0	0	0
0	0	1	1	1	0
0	0	1	1	1	0
0	1	1	1	0	0
0	1	1	1	0	0
0	1	1	0	0	0

A

0	1	0
1	1	1
0	1	0

S

(i)

1	1	1	1	1	1
1	1	0	0	0	1
1	1	0	0	0	1
1	0	0	0	1	1
1	0	0	0	1	1
1	0	0	1	1	1

(ii)

0	0	1	1	1	0
0	1	1	1	1	1
0	1	1	1	1	1
1	1	1	1	1	0
1	1	1	1	1	0
1	1	1	1	0	0

(iii)

0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	1	0	0
0	0	1	0	0	0
0	0	1	0	0	0
0	0	0	0	0	0

(iv)

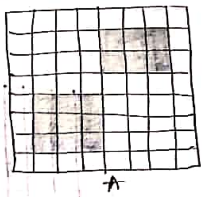
0	0	1	1	1	0
0	1	0	0	0	0
0	1		0	0	0
1	0	0		1	0
1	0	0		1	0
1	0	0	1	0	0

(v)

0	0	0	0	0	0
0	0	1	1	1	0
0	0	1	0	1	0
0	1	0	1	0	0
0	1	0	1	0	0
0	1	1	0	0	0

Quiz (Set-A)

1 (a)



A

B

(i) Erosion
(ii) Dilation
(iii) Opening
(iv) Closing

MORPHOLOGY - PART (II)

Maths & Prev years ques

YouTube (*) Given an 10X10 Img, use HMT

0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	0	0
0	1	1	1	1	1	1	1	0	0
0	1	1	1	1	1	1	1	0	0
0	1	1	1	1	1	1	0	0	0
0	1	1	1	1	1	0	0	0	0
0	1	1	1	1	0	0	0	0	0
0	1	1	1	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

A

$1 \rightarrow F$
$0 \rightarrow B$
$X \rightarrow$ Don't care

use the following SEs :

Corner Detection SEs :

B_1 ∟

X	1	X
0	1	1
0	0	X

B_2 ┘

X	1	X
1	1	0
X	0	0

B_3 ┐

X	0	0
1	1	0
X	1	X

B_4 ┌

0	0	X
0	1	1
X	1	X

$\Rightarrow$

I_1 =

0	0	0	0	0	0		0	0	0
0			0	0			0	0	0
0	0		0	0				0	0
	0	0	0				0		0
0			0						
				0		0			
0				0					
	1								

I_2 =

							1		
						1			
					1				
				1					
			1						

I_1, I_2, I_3, I_4

Scanned by CamScanner